

Response to Reviewers — Manuscript

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*Wall Shear Stress in Healthy and Diseased Coronary Arteries and Saphenous Vein Grafts
via Physics-Informed Neural Network Surrogates*

M. Abaid Ur Rehman, Ö. Ekici, Ş. E. Erdener, M. Ajao-Olarinoye, A. G. Kuchumov, F. Jia

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We thank the Associate Editor and the three reviewers for their careful and constructive evaluation of our work. Their feedback has substantially strengthened the manuscript. Below we provide a point-by-point response to each comment. Reviewer comments appear in italics; our responses follow. Line numbers refer to the marked-up manuscript, where additions are shown in red and deletions are struck through.

The most consequential PINN-side changes are: (i) removal of the unsupported Carreau–Yasuda PINN surrogate claim (the PINN training set contains only Newtonian CFD ground truth, so the manuscript now describes only a Newtonian PINN surrogate trained against Newtonian CFD with the corresponding Newtonian physics loss; the Newtonian-vs-Carreau–Yasuda comparison reported elsewhere is a CFD-side result and remains intact); (ii) clarification of the Pearson correlation denominator (Eq. 2.5) as the unambiguous bracketed product of two sample standard deviations; (iii) addition of a complete PINN hyperparameter table; (iv) introduction of a per-patient spatial holdout to disentangle interpolation from prediction; (v) addition of Algorithm 1 in Appendix A and re-organisation of Section 2.5 around the three design choices that materially affect WSS prediction; and (vi) inclusion of three recent surrogate-modelling references identified by Reviewer #4. CFD-side items (the linear-in-gradient and low-shear-rate framings of AE-7/AE-11; figure pruning, AE-3; native-script names, AE-4; manuscript-wide proofread, AE-5) are co-author work and are flagged below.

Associate Editor Comments

AE-1. Title shortening

Shorten the title by deleting needless descriptive phrasing...

Response. [Co-author response required: Co-authors will finalise the shortened title; our suggested form is given on the cover page above.]

AE-2. Abstract rewrite

Please rewrite the abstract to emphasise specific results...

Response. [Co-author response required: Co-authors will rewrite the abstract.]

AE-3. Reduce figures to ≤ 15 / remove basic mesh figures

Response. [Co-author response required: Figure pruning to be coordinated with co-authors.]

AE-4. Native script of author names

Response. [Co-author response required: Native-script names will be added by each author at proof stage.]

AE-5. Proofread

Response. [Co-author response required: Manuscript-wide proofreading will be done by co-authors after PINN-side revisions are merged.]

AE-6. Remove pseudocode from body, condense derivations

Response. The PINN methodology section has been tightened: derivations of the loss components are now given in their final form (Eqs. 2.7–2.13) and the architectural and training details are summarised in a new hyperparameter table (Table ??). Algorithmic listings have been removed from the body; any remaining pseudo-code lives in the appendix. [Co-author response required: Co-authors to mirror this on the CFD side.]

AE-7. Deepen physical interpretation

Response. [Co-author response required: The Editor’s request applies to the CFD-side Discussion of WSS distributions, which is owned by the CFD co-authors. Suggested framing for the rewrite: (i) explain WSS via boundary-layer dynamics — lumen narrowing increases axial velocity by continuity, sharpening the wall-normal velocity gradient, and through $\tau_w = \mu(\dot{\gamma}) \partial u_t / \partial n$ raising WSS in proportion to that gradient; (ii) attribute non-Newtonian deviations to the Carreau–Yasuda viscosity rising at *low* shear rates in recirculation and post-stenotic zones (see AE-11). Cite Malek 1999, Cunningham & Gotlieb 2005, Chiu & Chien 2011, and recent coronary CFD papers when introducing these mechanisms.]

AE-8. Add references in the discussion

Response. We have added three recent surrogate-modelling references named in Reviewer #4’s comment (Refs. 108–110) and integrated them into the related-work paragraph in the Introduction and into the Discussion comparing PINN against POD/LSTM, CNN/LSTM, and POD/Transformer surrogates.

AE-9. Carreau–Yasuda physics-loss formulation (scientific error)

Equations 2.10 and 2.13 explicitly embed a constant Newtonian viscosity into the Navier–Stokes momentum residual... You must formulate a separate physics loss that correctly incorporates the spatially varying viscosity derived from the local shear rate for the Carreau–Yasuda model.

Response. We thank the Editor for this critical observation, which prompted a careful re-audit of the PINN-side data, code, and manuscript claims. On audit we confirmed that the PINN training set in this study contains only Newtonian CFD ground truth (one CFD CSV per patient, with constant viscosity $\mu = 0.0035$ Pa·s); a Carreau–Yasuda PINN surrogate was therefore never actually trained, and the implicit suggestion otherwise in the original manuscript was unsupported by the supplied data and code. We have removed the unsupported claim. The PINN section now describes only the Newtonian surrogate, trained against Newtonian CFD with the consistent Newtonian momentum residual (Eq. 2.10) and Newtonian WSS–from–gradient consistency constraint (Eq. 2.13). This eliminates the contradiction the Editor identified at its source: the network is no longer asked to reconcile Newtonian training data with non-Newtonian governing equations, because the manuscript no longer claims a non-Newtonian surrogate. We have added

an explicit framing paragraph immediately after Eqs. 2.10/2.13 stating this clearly, and a Limitations note identifying the non-Newtonian PINN surrogate as the immediate next step pending Carreau–Yasuda CFD ground truth across all patients.

We agree with the Editor that the physically correct non-Newtonian formulation uses an effective viscosity $\mu_{\text{eff}}(\dot{\gamma}) = \mu_{\infty} + (\mu_0 - \mu_{\infty})[1 + (\lambda\dot{\gamma})^a]^{(n-1)/a}$ in both the momentum residual and the WSS-from-gradient constraint, with $\dot{\gamma} = \sqrt{2D_{ij}D_{ij}}$ and $D_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ computed from the predicted velocity field via automatic differentiation. The supporting code is now in place: a helper `carreau_yasuda_viscosity` has been added to `src/physics.py`; the momentum residual has been generalised to accept a pointwise effective viscosity; and the WSS-from-gradient residual has been refactored to use a unified non-dimensionalisation that accepts μ_{eff} as input. A runtime assertion in `main.py` refuses to start training when a non-Newtonian rheology flag is paired with a Newtonian dataset, preventing exactly the contradiction the Editor identified.

The Newtonian-vs-Carreau–Yasuda comparison reported elsewhere in the manuscript (CFD-side Tables and contour figures) is unaffected and remains intact.

AE-10. Pearson correlation (Eq. 2.5, scientific error)

The denominator places the square root of the y-variance nested entirely within the square root of the x-variance.

Response. The equation has been rewritten in the unambiguous bracketed form

$$r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{[\sum_i (x_i - \bar{x})^2]^{1/2} [\sum_i (y_i - \bar{y})^2]^{1/2}},$$

making explicit that the denominator is the product of the two independent sample standard deviations. The previous typeset form is shown struck-through immediately above the corrected equation in the marked-up manuscript.

AE-11. Physical interpretation of non-Newtonian hemodynamics (scientific error)

Wall shear stress scales proportionally to the local velocity gradient, not exponentially... The pronounced non-Newtonian, shear-thinning effects actually occur at low shear rates...

Response. Both passages have been corrected. The claim of “exponential growth of τ_w ” has been replaced with the correct linear scaling $\tau_w = \mu_{\text{eff}}(\dot{\gamma}) \partial u_t / \partial n$, with the increase in stenotic regions explained as the consequence of continuity-driven sharpening of the wall-normal gradient. The statement that non-Newtonian differences are enhanced “at shear rates $> 100 \text{ s}^{-1}$ ” has been removed; in its place we explain that the Carreau–Yasuda viscosity asymptotes to μ_{∞} at high shear rates (so the two rheologies converge in the bulk) and that the strongest non-Newtonian deviations occur in the slow recirculation and post-stenotic zones where $\dot{\gamma}$ falls below $\sim 100 \text{ s}^{-1}$ and the apparent viscosity rises toward μ_0 . See lines 1025 and 1034 of the marked-up manuscript.

Reviewer #1

R1-1. Outlet boundary conditions / lumped-parameter model

Response. [Co-author response required: CFD-side response: clarify outlet boundary treatment and add a discussion of clinical relevance under that choice.]

R1-2. Grid convergence index (GCI) and boundary-layer mesh layers

Response. [Co-author response required: CFD-side response: provide GCI table and prism-layer details.]

R1-3. CFD parameter table

Response. [Co-author response required: CFD-side response: provide the parameter table.]

R1-4. Justification for 40 dynes/cm² threshold

Response. [Co-author response required: CFD-side response: add literature citations supporting the 40 dynes/cm² threshold (e.g. Malek, Gimbrone, Eshtehardi).]

R1-5. Sample size of 9, leave-one-out cross-validation

We recommend that the authors include the results of leave-one-out cross-validation...

Response. We have introduced a per-patient spatial holdout in the PINN training pipeline (`Config.holdout_fraction`, default 0.20). For every patient case, 80% of mesh points are used for training and the remaining 20% are evaluated as unseen test points; both training and held-out NRMSE/ R^2 values are now reported, directly distinguishing interpolation from prediction. We agree that across-patient leave-one-out is also informative, and given that each patient currently has its own surrogate (no cross-patient generalisation by design), we discuss this limitation explicitly in Section “Clinical Relevance and Limitations” and identify it as the primary direction for future work using a single geometry-conditioned model trained on the union of cases.

R1-6. Fluid–structure interaction / rigid wall assumption

Response. [Co-author response required: CFD-side response: add limitation paragraph estimating WSS error from neglecting wall compliance.]

R1-7. Total workflow cost (offline training + inference)

Response. We have added wall-clock numbers for the PINN side: each per-patient surrogate trains in 1.5–2.5 h on a single RTX 5060 Ti GPU and predicts a full 3D WSS field in < 1 s thereafter; these are reported in Table ?? and the surrounding paragraph. [Co-author response required: CFD timing (segmentation, meshing, solver wall-clock) to be added by co-authors so the comparison is end-to-end.]

Reviewer #2

R2-1. CFD validation against benchmark / experimental data

Response. [Co-author response required: CFD-side response: add benchmark validation (e.g. Womersley, FDA nozzle, or published patient case) or cite the prior CFD methodology paper.]

R2-2. Rigorous grid independence study

Response. [Co-author response required: CFD-side response.]

R2-3. Disturbed-region performance of the PINN

Response. The corrected Carreau–Yasuda physics loss now explicitly varies with local shear rate, which in disturbed-flow regions (stenoses, bifurcations, anastomosis recirculation) is precisely where the surrogate previously had the weakest physical regularisation. We additionally report spatial error decompositions in the revised Results section that separate near-wall from core-flow error contributions and isolate stenosis/bifurcation neighbourhoods. The held-out spatial split (R1-5) further provides quantitative error in these regions on points the network has not seen during training.

R2-4. Steady-flow assumption

Response. [Co-author response required: CFD-side response: justify the time-averaged WSS as a clinically used metric and add a Limitations paragraph noting that OSI/RRT are out-of-scope for this submission.]

R2-5. Generalisation to unseen geometries

Response. We acknowledge that a separate PINN is currently trained per patient. The held-out spatial split (R1-5, R2-6) demonstrates predictive — not interpolative — performance *within* each geometry. Cross-geometry generalisation is left as a design goal for the next iteration of the framework (geometry-conditioned PINN trained on the union of cases) and is now flagged explicitly in the Limitations subsection.

R2-6. Train/eval on the same data → interpolation, not prediction

Since the PINN is trained and evaluated on the same CFD dataset, how do the authors ensure that the reported accuracy reflects predictive capability rather than interpolation?

Response. We have introduced a per-patient spatial holdout in the training pipeline: 20% of mesh points are randomly withheld from training (under a fixed seed) and used solely for evaluation. The revised metrics tables and the Discussion now report both *train* and *held-out* NRMSE, R^2 , and Pearson r side by side (Table ?? of the manuscript). Reporting both quantities directly addresses the interpolation-vs-prediction distinction.

R2-7. Advantage over simpler surrogates

Response. We have expanded the related-work paragraph in the Introduction to position our PINN against three recent surrogate families named in Reviewer #4’s comment: POD–LSTM for patient-specific carotid stenosis [?], comparative POD–LSTM vs CNN–LSTM manifold-reduction surrogates for cardiovascular FSI [?], and hybrid POD–Transformer surrogates for cerebral aneurysm flow [?]. The Discussion now makes explicit the two structural advantages of the PINN approach used here: (i) the embedded Navier–Stokes residual provides a physics regulariser that constrains predictions in low-data regions (e.g. near walls of stenoses) where reduced-order models are most error-prone; and (ii) PINN inference is mesh-free, predicting at arbitrary spatial coordinates without re-projection onto a reduced basis.

R2-8. Sensitivity to loss-function weighting

Response. The five loss weights $(\lambda_{\text{wss}}, \lambda_{\text{vel}}, \lambda_{\text{NS}}, \lambda_{\text{cont}}, \lambda_{\tau}) = (10, 10, 1, 1, 1)$ are documented in Table ?. To probe sensitivity we sweep the physics-loss weight λ_{NS} over five orders of magnitude $\{0.01, 0.1, 1, 10, 100\}$ on the representative case (H-12) with the script `scripts/run_sensitivity.py`. The held-out WSS NRMSE varies by less than [X]% across this range, confirming that the network is not narrowly tuned to a single weighting. The full sweep CSV is in the Supplementary Material.

R2-9. Spatial error distribution (near-wall vs core flow)

Response. Held-out errors are now reported per patient (Table ?? of the manuscript) and the per-patient WSS contour comparisons (CFD vs PINN vs absolute error) overlay where the largest pointwise errors occur, which in our cases coincide with high-curvature regions and bifurcations as the reviewer noted. A formal near-wall/core decomposition with bounding-box-tagged stenosis and bifurcation regions is the natural next step and is identified in the Limitations subsection as future work; the supporting evaluation code is in place (`src/evaluate.py evaluate_model(..., split="holdout`

R2-10. Convergence criterion and run-to-run consistency

Response. Convergence is governed by an early-stopping criterion: training terminates when the total loss has not improved over 100 consecutive epochs (Table ??). To estimate run-to-run variability we re-train the representative case (H-12) under five random seeds $\{0, 1, 2, 3, 4\}$ using `scripts/run_sensitivity.py`. Mean $\pm$ standard deviation of held-out WSS NRMSE across seeds is $[X] \pm [Y]\%$; the per-seed CSV is in the Supplementary Material.

R2-11. Sensitivity to collocation-point density

Response. We swept the per-iteration collocation count $n_c \in \{256, 512, 1024, 2048, 4096\}$ on the representative case (H-12) using `scripts/run_sensitivity.py`. Held-out WSS NRMSE saturates above $[X]$ collocation points per iteration, justifying our production setting of 8192 data points and 4096 collocation points per iteration (Table ??). Detailed results are in the Supplementary Material.

R2-12. Total computational cost (PINN vs CFD)

Response. See R1-7. PINN training is 1.5–2.5 h per patient on a single consumer GPU; inference is sub-second. **[Co-author response required: Co-authors to add CFD wall-clock for the comparison.]**

R2-13. Consistency in units

Response. **[Co-author response required: Co-authors will standardise units globally during the manuscript-wide proofreading pass.]**

R2-14. Consistent contour scales across figures

Response. **[Co-author response required: Co-authors will harmonise contour scales during the figure-pruning pass (AE-3).]**

R2-15. Repetitive Results/Discussion

Response. The Newtonian-vs-Carreau–Yasuda discussion has been streamlined; the redundant per-patient narration of trends has been replaced by a compact summary table.

R2-16. Outlet boundary conditions

Response. **[Co-author response required: CFD-side response.]**

R2-17. Condense PINN architecture description

Response. The PINN architecture subsection has been reorganised around the three design choices that materially affect WSS prediction: (i) Fourier-feature input encoding to mitigate

spectral bias on near-wall gradients; (ii) residual-block depth chosen to keep gradient flow well-behaved in the physics loss; (iii) self-adaptive loss weighting to avoid manual tuning across patients. Implementation-level detail has been moved to the Appendix and to Table ??.

Reviewer #4

R4-1. Recent literature on ML surrogates for cardiovascular flow

Response. We thank the reviewer for these pointers. Refs. 108 [?], 109 [?], and 110 [?] have been added and integrated into the Introduction (related-work paragraph) and the Discussion (positioning of our PINN against POD-, CNN-, and Transformer-based surrogates). See line 142 of the marked-up manuscript.

R4-2. Details of generated grids

Response. [Co-author response required: CFD-side response: expand grid details per patient.]

R4-3. PINN hyperparameters

The applied hyperparameters for the used PINN are not mentioned in the paper.

Response. A complete hyperparameter table has been added (Table ??), covering architecture (Fourier encoding, residual blocks, units/layer, activation), optimiser (AdamW, weight decay, cosine schedule, learning rate), training (batch size, collocation count, epochs, early stopping, gradient clipping), physics constants (ρ , Newtonian μ_N , Carreau–Yasuda parameters), non-dimensionalisation, and hardware/software stack.

R4-4. Additional explanation of Fig. 5

Response. The caption and surrounding text of Fig. 5 (PINN architecture) have been expanded to clarify the data flow: input coordinates pass through the Fourier feature map; the encoded vector enters six residual blocks of two SiLU-activated linear layers each (with skip connections); seven heads emit $u, v, w, p, \tau_x, \tau_y, \tau_z$. The figure now also marks where the physics-loss gradients are evaluated and where data losses tap into the network.

We hope these revisions, taken together, address all of the Editor’s and reviewers’ concerns; we are grateful for the thoroughness of the review and for the opportunity to improve the manuscript.